Memory Performance: Factorial ANOVA

Two-way ANOVA analysis in R

Executive Summary

This analysis tested whether memory scores differed across three spacing intervals (long, medium and short), two material formats, or the interaction between those factors. The balanced dataset contained 180 observations, with 30 observations in each of the six experimental conditions.

A Type II two-way ANOVA found no statistically significant main effect of spacing, no significant main effect of material format and no significant interaction. All partial eta-squared values were below 0.01, showing that the experimental conditions explained less than 1% of the variation in memory scores. The visual pattern contains a line crossing, but the differences occur within a narrow score range and are not statistically reliable.

180	6	0.0074	0
Total observations	Conditions	Largest effect size	Significant effects

Research Question

Do spacing interval, material format, or their combination meaningfully change memory performance?

Experimental Design

Factor	Levels	Role
Spacing interval	Long, medium, short	Independent variable
Material format	Visual, verbal	Independent variable
Memory score	Continuous score	Dependent variable

The design is a balanced 3 x 2 factorial structure. A factorial ANOVA is appropriate because it estimates both main effects and whether the effect of one factor depends on the level of the other factor.

Analytical Workflow

- Imported the simulated memory dataset into R and converted spacing and material into categorical factors.
- Calculated group-level sample sizes, means, standard deviations and medians.
- Checked homogeneity of variance with Bartlett's test and inspected normality using a histogram and Shapiro-Wilk test.
- Fitted a linear model with spacing, material and their interaction, then evaluated it with a Type II ANOVA.
- Calculated partial eta squared to separate statistical significance from practical importance.

- Used an interaction plot and grouped boxplots to compare the visual pattern with the model results.

Assumption Assessment

Assumption	Result	Interpretation
Homogeneity of variance	Bartlett $p = .25$	Met; group variances were not detectably different
Normality	Shapiro-Wilk $W = .91$, $p < .001$	Violated; scores were not perfectly normal
Balance and group size	$n = 30$ per cell	Balanced moderate-sized groups increase ANOVA robustness

Interpretation of assumptions

The normality test was statistically significant, so the data should not be described as normally distributed. However, the design was balanced and each cell contained 30 observations. This makes the ANOVA reasonably robust, while the violation remains an explicit limitation rather than something to ignore.

ANOVA Results

Effect	F statistic	p-value	Partial eta squared	Conclusion
Spacing interval	$F(2,174) = 0.646$	.525	.0074	Not significant; negligible effect
Material format	$F(1,174) = 0.144$	.705	.0008	Not significant; negligible effect
Spacing x material	$F(2,174) = 0.646$	.525	.0074	No reliable interaction

None of the p-values approached the conventional .05 threshold. The effect sizes reinforce this result: even the largest effects explain only about 0.74% of the observed variance. The analysis therefore supports a null finding in both statistical and practical terms.

Effect-Size Perspective

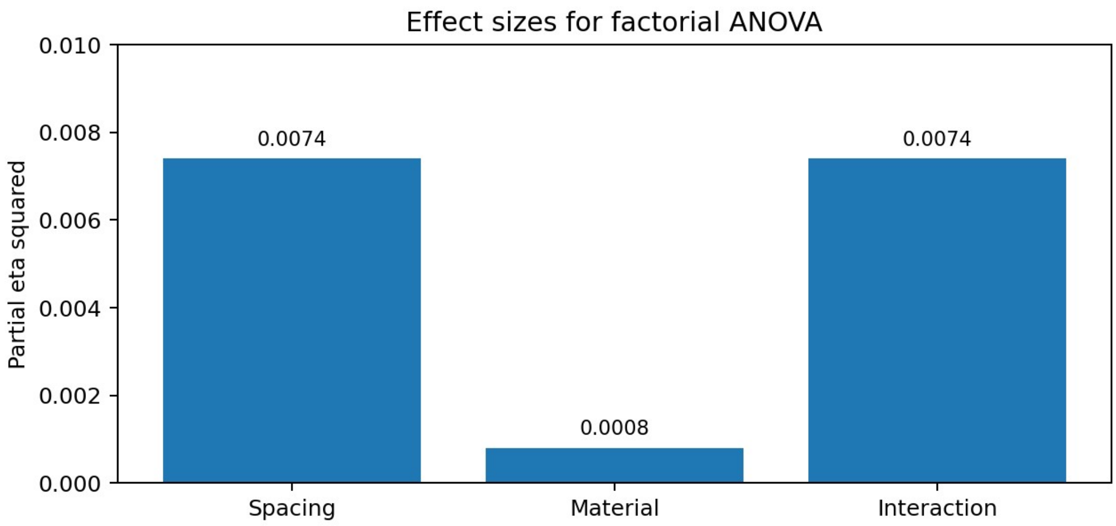

Figure 1. Partial eta-squared values for the two main effects and their interaction.

Visual interpretation

All bars remain below 0.01. The chart makes the practical result clear: the experimental factors contributed very little explanatory power, even before considering whether their p-values were statistically significant.

Interaction Plot

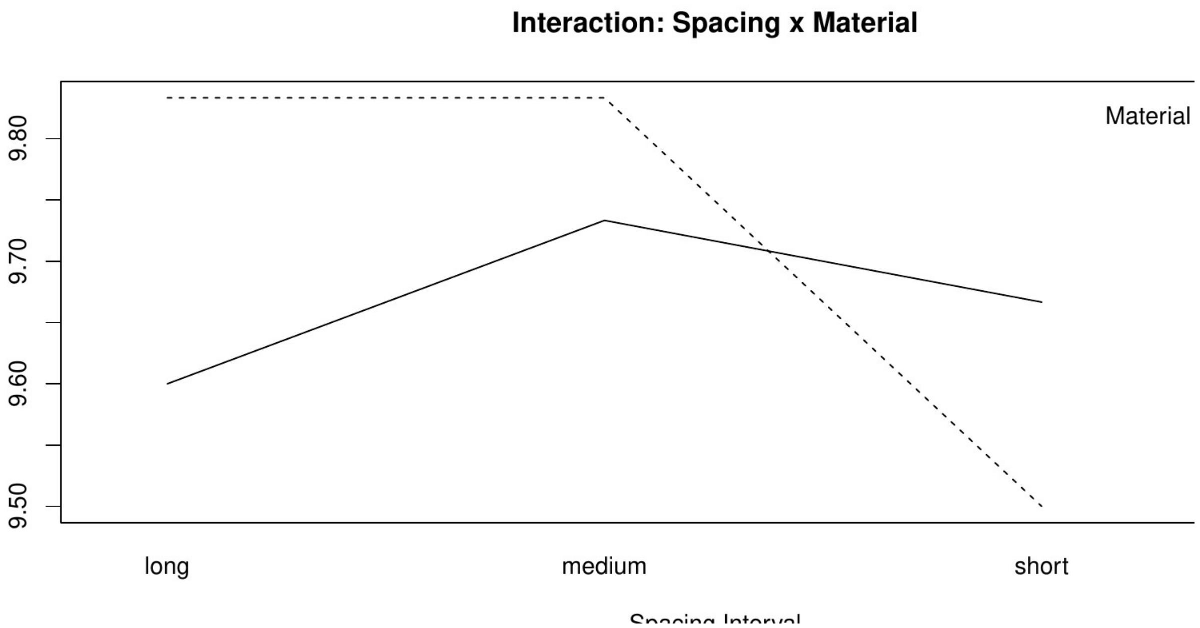

Figure 2. Mean memory scores across spacing intervals for the two material formats.

Visual interpretation

The lines are not parallel and cross between the medium and short conditions, which can initially look like an interaction. However, all condition means lie within a narrow band of roughly 9.5 to 9.8. The apparent crossing therefore represents a small fluctuation rather than a stable, meaningful change in how spacing affects the two material formats. The non-significant interaction test and partial eta squared of .0074 confirm that the visual pattern should not be overinterpreted.

Conclusion

The study found no evidence that memory performance was meaningfully affected by spacing interval, material format or their interaction. The most defensible conclusion is not that the factors can never matter, but that this dataset does not provide evidence of a meaningful effect.

Limitations and Next Steps

- The Shapiro-Wilk test indicated non-normality, although the balanced design reduces sensitivity to the violation.
- Scores were tightly clustered, which may have limited the ability to detect meaningful differences.
- A future study could use a more sensitive memory measure, verify the manipulation strength and report confidence intervals for each group mean.
- Replication with an independently collected sample would be required before drawing general conclusions about learning design.